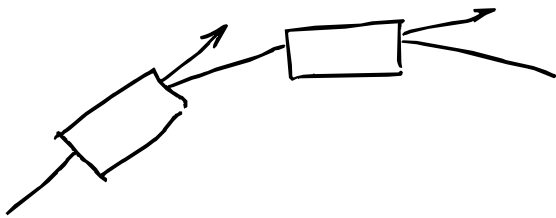


Radiation Distributions + Damping

Yesterday, found that radiation emitted depends on acceleration.

→ Use that fact to shape/amplify the radiation

BEND RADIATION



SLIDE 2

→ properties of radiation limited

→ bends must meet machine requirements.

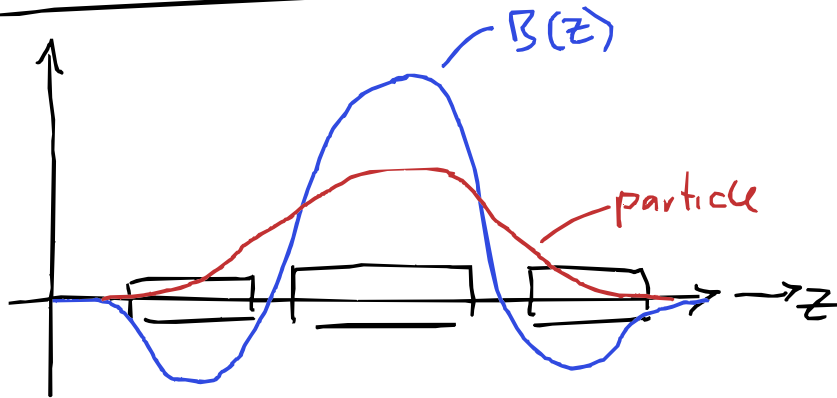
But you can replace some (or all) of the bends with "super bends"

→ increase field (SC → 5-6T)

∴ push $\omega_c \uparrow$

$$E_c = 0.665 E^2 [\text{GeV}] B [\text{T}]$$

WAVELENGTH SHIFTER

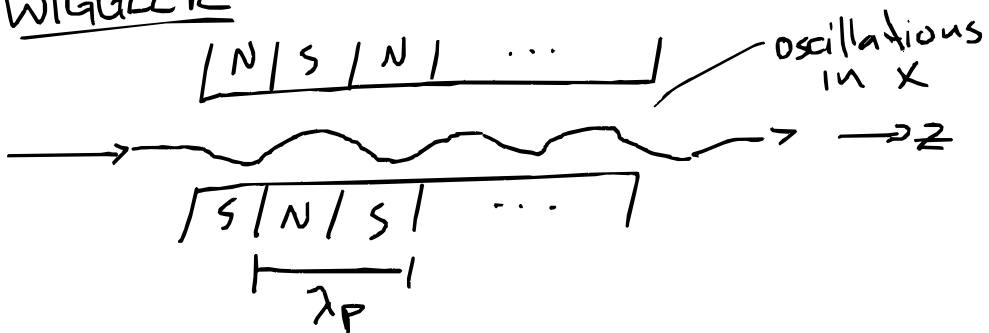


→ no net deflection

→ $\frac{1}{2}$ period wiggler

SLIDE 3

WIGGLER



$$B_y = B_0 \sin\left(\frac{2\pi z}{\lambda_p}\right) \quad (\text{for short period})$$

$$\Rightarrow \theta_{\max} = \frac{B_0}{B_p} \int_0^{\lambda_p/4} \sin\left(\frac{2\pi z}{\lambda_p}\right) dz$$

$$= \frac{B_0}{B_p} \frac{\lambda_p}{2\pi}$$

Define the wiggler strength parameter

$$K = \gamma \theta = \frac{ecB_0}{mc^2} \frac{\lambda_p}{2\pi}$$

$$K = 0.934 B_0 [T] \lambda_p [cm]$$

Wigglers have $K \gg 1$

Recall: sync. power: $P_\gamma = \frac{2}{3} \gamma_c mc^3 \frac{B^4 \gamma^4}{\rho^2}$

$$\text{Wiggler Power: } \langle P_\gamma \rangle = \frac{1}{3} \gamma_c mc^2 c \gamma^2 K^2 \frac{4\pi}{\lambda_p^2}$$

$$\rightarrow \langle P_\gamma [W] \rangle = 632.7 E^2 B_0^2 I L_u$$

beam
current

$N_p \lambda_p$
= length

UNDULATORS

$k \ll 1 \rightarrow$ undulator

$\Rightarrow$ give weak deflections to the particles

$$\theta_{\max} < \pm 1/\gamma$$

$\therefore e^-$ undergoes $\sim$ sinusoidal oscillations

If we go to the boosted frame of the e^- (in z direction)

$\Rightarrow$ transverse motion looks like dipole motion

$\rightarrow$ emits radiation at length contracted period of the undulator

SLIDE 5

In the lab frame

$\rightarrow$ radiation spectrum broadens

+ radiation become forward directed within $\theta < \pm 1/\gamma$

As the undulator strength increases $\rightarrow$ transverse motions become relativistic

$\Rightarrow$ get harmonics in the emission

$$\lambda_1 = \frac{\lambda_p}{2\gamma^2} \left(1 + \frac{1}{2} k^2 + \gamma^2 \theta_{\text{obs.}}^2 \right)$$

for flat undulator.

SLIDE 6

More on this tomorrow

Leave with this slide to show
main distinctions b/w

Bend vs Undulator vs Wiggler
radiation

SLIDE 7

Review of Long. Dynamics (in ring)

synch. freq: $\Omega^2 = \frac{\omega_{\text{rev}}^2 h \eta_c V_0 \cos \phi_s}{2\pi \beta c \rho_0}$

Rev. freq $\omega_{\text{rev}} \rightarrow \omega_{\text{rf}} = h \omega_{\text{rev}}$

slip factor $\eta_c = \left(\frac{1}{\gamma^2} - \alpha_c \right)$

Mtm compaction $\alpha_c = \Delta C / C_0 / \Delta P / P_0$

transition Energy $\gamma_{\text{tr}} = \frac{1}{\sqrt{\alpha_c}}$

Synch. phase ϕ_s

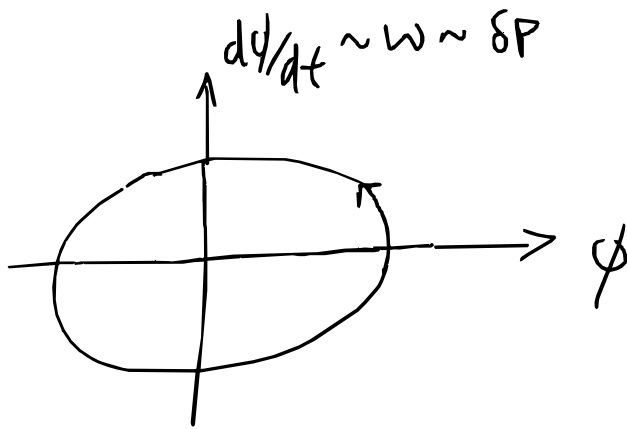
Max cavity V $V_0 \rightarrow V(\phi) = V_0 \sin \phi$

Synchrotron tune $\nu_s = \frac{\Omega}{\omega_{\text{rev}}}$

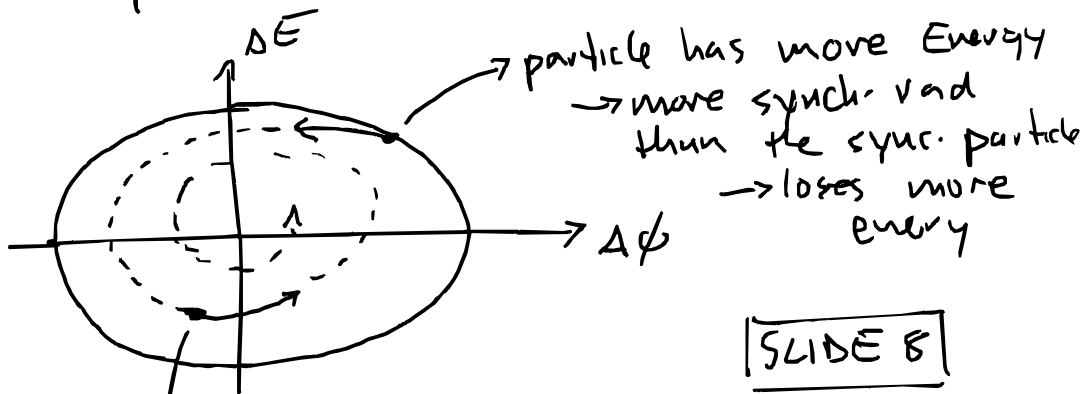
$\hookrightarrow \nu_s$ tends to be $\ll 1$

Eqn of Motion $\ddot{\phi} + \Omega^2 \phi = 0$

$\hookrightarrow$ Describes oscillations in ϕ and $\dot{\phi}$ phase space.



How does this evolve when we account for synchrotron radiation?



⇒ spirals inwards ⇒ damping of the long. oscillation.

For a particle on ref. energy
Total energy loss to SR is
the integration of radiated power:

$$U_0 = \oint P_r dt = \frac{1}{c} \oint P_r ds$$
$$= \frac{C_r}{2\pi} E_0^4 \int \frac{1}{\rho^2} ds$$

Define: Second Synchrotron radiation integral

$$I_2 = \int \frac{1}{\rho^2} ds$$

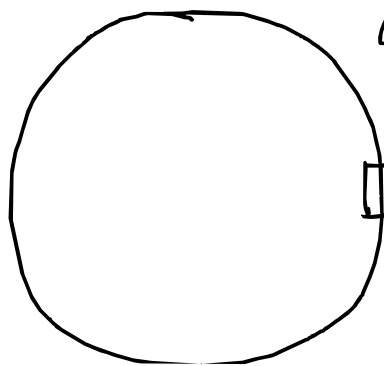
$$\rightarrow U_0 = \frac{C_r}{2\pi} E_0^4 I_2$$

There are 5 radiation
integrals.

See Lee see 4, II, 8

Damped Eq'n of Motion

Lee 4.II



$$L = Z\pi\rho$$

$$\square V = V_0 \sin\phi = V_0 \sin \omega_{rf}(\tau + \Delta\tau)$$

$$\underline{U_0 = eV_0 \sin\phi_s}$$

Consider motion in ring, with energy loss to SR, $\underline{U_0}$

With RF cavity providing the same energy back (at sync. phase), $\underline{eV_0 \sin\phi_s}$

$$\text{Recall: } \alpha_c = \frac{\Delta L/L}{\Delta p/p} \quad (\text{intm compaction})$$

The diff in arrival time of a particle off energy is

$$\Delta\tau = \frac{\Delta L}{c} = \alpha_c \frac{L}{c} \frac{\Delta E}{E} = \alpha_c T_0 \frac{\Delta E}{E}$$

$$\text{Make a derivative: } \left[\frac{d\tau}{dE} = \frac{\Delta\tau}{T_0} = \alpha_c \frac{\Delta E}{E} \right]$$

In one revolution, the particle
loses $U(E)$ by sync. rad.
gains $eV(\tau)$ from the RF cavity

$\therefore$ The net change in energy is:

$$\frac{d(\Delta E)}{dt} = \frac{eV(\tau) - U(E)}{T_0}$$

But what is $U(E)$?

Write it in terms of an offset from
the synchronous particle:

$$U(E) = U_0 + W \Delta E$$

$\nearrow$
SR of synchron. particle

$$\nwarrow W = \left. \frac{dU}{dE} \right|_{E=E_0}$$

So W tells us how the SR energy
radiated changes, based on the
energy of that particle.

(in ref. to the sync. particle of course)

A reminder: This setup is a storage ring, so no net change in energy of the synchronous particle.

$$U_{rf} = eV(\tau) = U_0 + e \frac{dV}{d\tau} \tau$$

loss to SR

$$eV_0 \sin(\omega_{rf} \tau)$$

$$\dot{V} = \omega_{rf} V_0 \cos(\omega_{rf} \tau)$$

Take our equation from before

$$\frac{d(\Delta E)}{dt} = \frac{eV(\tau) - U(E)}{T_0}$$

$$= \frac{1}{T_0} \left(e\dot{V}\tau + \underbrace{U_0 - U(E)}_{-W\Delta E} \right)$$

$$\Rightarrow \boxed{\frac{d(\Delta E)}{dt} = \frac{1}{T_0} (e\dot{V}\tau - W\Delta E)}$$

Now, come back to our equation with the momentum compaction term:

$$\frac{d\tau}{dt} = \alpha_c \frac{\Delta E}{E}$$

Differentiate w. time:

$$\frac{d^2\tau}{dt^2} = \frac{\alpha_c}{E} \frac{d(\Delta E)}{dt}$$

Plug in our previous result.

$$\frac{d^2\tau}{dt^2} = \frac{\alpha_c}{E T_0} (e V \tau - W \Delta E)$$

Rearranging:

← ω_s contains the -

$$\ddot{\tau} + 2 \left(\frac{W}{2T_0} \right) \left(\frac{\alpha_c \Delta E}{E} \right) - \left(\frac{\alpha_c e V}{T_0 E} \right) \tau = 0$$

$$\rightarrow \boxed{\ddot{\tau} + 2 \alpha_E \dot{\tau} + \omega_s^2 \tau = 0}$$

This is just the eqn for a damped harmonic oscillator, with damping coefficient α_E .

For $\alpha_E \ll \omega_s$

$$\Rightarrow \tau(t) = A e^{-\alpha_E t} \cos(\omega_s t - \theta_0)$$

$$\alpha_E = \frac{\omega}{Z \tau_0}$$

To get α_E , we need to calculate

$$\omega = \left. \frac{d\omega}{dE} \right|_{E=E_0}$$

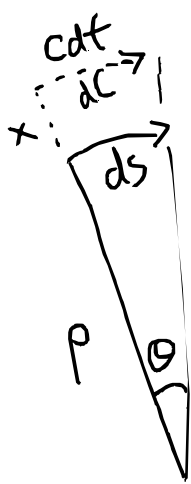
Recall $\rightarrow \omega$ determines how the SR changes with the energy of that particle. (w.r.t. to synch. particle)

$$U = \oint P_T dt = \oint P_T \frac{dt}{ds} ds$$

Note that we now have to consider how x and Energy interact as E changes, so does x

as related by dispersion $x = D \frac{\Delta E}{E_0}$

Recall - particle w. higher energy takes longer time to orbit



$$ds = \Theta p \rightarrow \text{arc of on-mtn } e^-$$

$$dc = \Theta(p+x) \rightarrow \text{arc of higher } E e^-$$

$$\rightarrow \frac{ds}{p} = \frac{dc}{p+x} = \Theta$$

$$\frac{dc}{ds} = \frac{p+x}{p}$$

$$\rightarrow \frac{dt}{ds} = \frac{1}{c} \left(1 + \frac{x}{p}\right)$$

Then the SR loss per rev. is:

$$U = \frac{1}{c} \oint P_r \left(1 + \frac{D}{p} \frac{\Delta E}{E_0}\right) ds$$

After some math...

$$\boxed{\alpha_E = \frac{1}{Z T_0} \frac{dU}{dE} = \frac{U_0}{Z T_0 E} (Z + \bar{D})}$$

where $\bar{D}$ is the damping
partition number

$\bar{D}$ is a property of the lattice configuration. (D in Lec)

→ Radiation integrals!

Note: emission is QM process

→ SR emitted as discrete energy photons
w. statistical fluctuations.

∴ there is a limit to the damping
by SR.

and SR $\Rightarrow$ energy spread
in equilibrium.

Last note:

What is a damping ring?

[SLIDE 9]

SR emitted $\sim$ in direction of mfm vector

